

# Derivative

$$f(x) = \frac{1}{3\sqrt{2+x}}$$

$$f(x) = \frac{1}{3} \cdot \frac{1}{\sqrt{2+x}}$$

$$\left(\frac{1}{\sqrt{2+x}}\right)' = \frac{u'}{u^2} = \frac{\frac{1}{2\sqrt{2+x}}}{\sqrt{2+x}^2} = \frac{\frac{1}{2\sqrt{2+x}}}{2+x}$$

$$\text{then: } \frac{1}{2\sqrt{2+x}(2+x)} \cdot \frac{1}{3}$$

$$\text{So, } f'(x) = \frac{1}{6\sqrt{2+x}(2+x)}$$

